The `XMLHttpRequest` object exposes two methods for reading the response headers: `getResponseHeader` and `getAllResponseHeaders`. Same-origin requests can use these methods to read headers from the response. But cross-origin requests have limitations on which response headers can be viewed by the client. By default, only a few response headers are visible to clients on cross-origin requests (these are called simple response headers; see the sidebar "Simple response headers" later in the chapter for more details). The server must give its permission to read any of the other response headers. Let's modify the sample app to read one of these headers.

5.2.1 Reading a response header

The X-Powered-By header is often used by servers to highlight the underlying technology that powers the server. Values for this header could be `ASP.NET` (for a server running on ASP.NET) or `PHP/5.2.4` (for a server running on PHP). By default, Express sets the X-Powered-By header to `Express`. Figure 5.8 shows where to find the X-Powered-By header in the API response.

Because the X-Powered-By header exists on every response, let's display it in the sample app. The following listing reads the header using `XMLHttpRequest`'s `getResponseHeader` method. If the header exists, it's displayed on the page.

Listing 5.5 Reading the X-Powered-By response header

```
xhr.onload = function() {
  var data = JSON.parse(xhr.responseText);
  var elem = document.getElementById('output');

  var xPoweredBy = xhr.getResponseHeader('X-Powered-By');
  if (xPoweredBy) {
    var xpbDiv = document.createElement('div');
    xpbDiv.className = 'post';
    xpbDiv.innerHTML = 'X-Powered-By: ' + xPoweredBy;
    elem.appendChild(xpbDiv);
  }

  for (var postId in data) {
```

Reads X-Powered-By header from HTTP response

Displays header on page

If you reload the page, you'd expect to see the X-Powered-By header displayed on the page. Since this is a cross-origin request, the header comes up empty. (Some browsers such as Chrome even display a helpful error message in the console log: `Refused to get unsafe header "X-Powered-By"`). So let's update the server to give you access to that header.

5.2.2 Adding response header support

The server needs to specify that it's okay for the client to read the X-Powered-By header. The server does this by using the Access-Control-Expose-Headers header. The Access-Control-Expose-Headers header contains a list of headers that the client code can read.

This shouldn't be confused with the Access-Control-Allow-Headers header that was covered in chapter 4. While they are similar, there is no connection between the two.